

FEATURE ENGINEERING & DATA PREPROCESSING REPORT

1. Introduction

This project focuses on preprocessing raw data and transforming it into a machine learning-ready format. The goal is to improve data quality using encoding, scaling, and feature engineering techniques.

2. Dataset Description

The dataset contains a mix of numerical and categorical features. It includes multiple independent variables and a target variable used for prediction.

- Number of rows: XXX
 - Number of columns: XXX
 - Target variable: (mention your target column name)
-

3. Data Preprocessing Steps

◆ Handling Missing Values

Missing values were checked and treated using mean/median imputation for numerical features.

Encoding Categorical Variables

Categorical columns were converted into numerical format using:

- One-Hot Encoding (for nominal data)
 - Label Encoding (for ordinal data)
-

Feature Scaling

Numerical features were standardized using StandardScaler to bring all values to a similar range.

4. Feature Engineering

New meaningful features were created from existing data:

- Total Score = sum of multiple numeric columns
- Average Score = mean of selected features
- Interaction features (if applicable)

These features help improve model performance by providing better patterns.

5. Why Preprocessing is Required

Raw data cannot be directly used for machine learning models due to:

- Missing values
- Categorical variables
- Different feature scales

Preprocessing ensures:

- Better accuracy
- Faster model convergence
- Improved model performance

6. Conclusion

The dataset was successfully cleaned, transformed, and converted into a model-ready format using preprocessing and feature engineering techniques.